

paired test conditions (same target cue, labels, noise)

pixel spectra $x_p \in \mathbb{R}^S$

iid
context = label

reversed
context = -label

context-random
context independent

neighbours: contextual cue
 $\beta(y_p + \tau\eta)v_d$, large amplitude

each pixel

encoder
 $W \in \mathbb{R}^{K \times S}$
shared, per pixel

$z_p = Wx_p$

spatial head
width M , ReLU
reads the encoded
neighbourhood
rate $\kappa\eta$

logit F_p

$\mathcal{L} = \overline{\log(1 + e^{-y_p F_p})}$
base rate η , declared block
multipliers; stop at matched fit L^*

centre: spectral cue
 $\alpha y_p u + \text{isotropic noise}$